Technical University of Denmark

Time Series Analysis 02417

Spring 2026

Assignment 1

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Date
March 2, 2026

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1 Plot data

1.1 Time variable x

We work with monthly data from Statistics Denmark (table `BIL54`). The variable of interest is *total* ("Drivmidler i alt"), i.e. the number of motor driven vehicles registered in Denmark.

A continuous time variable x_t is defined such that January 2018 is $x_1 = 2018$ and each next month adds $\frac{1}{12}$:

$$x_t = 2018 + \frac{t-1}{12}, \qquad t = 1, 2, \ldots$$

The output is scaled to millions of vehicles for readability.

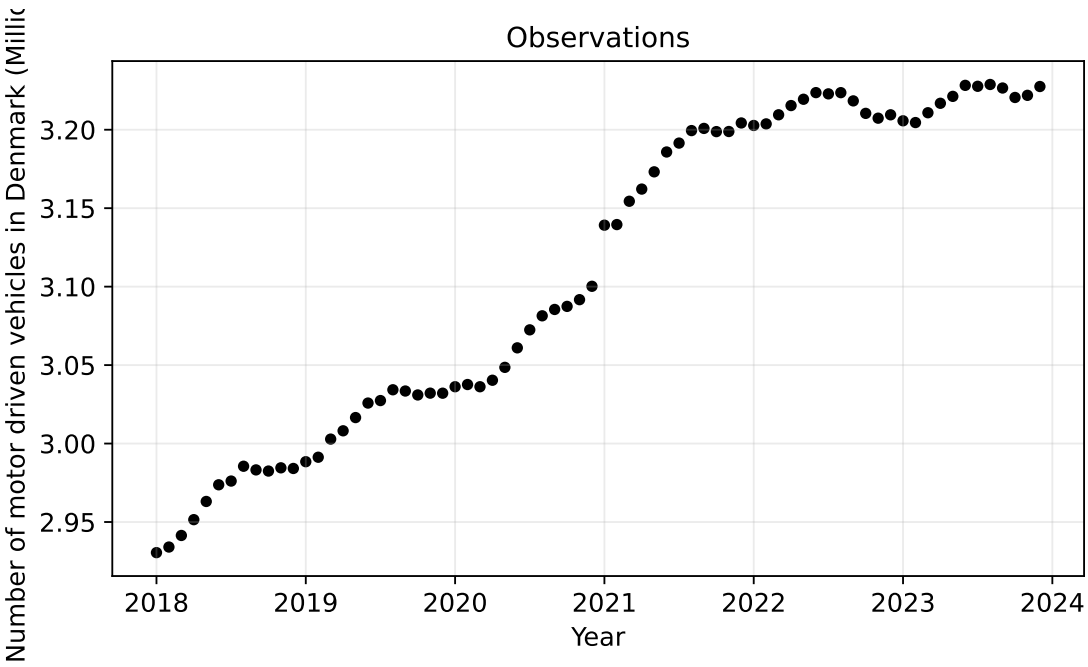

Figure 1: Number vehicles in registered in Denmark from 2018 to 2023.

1.2 Description of the time series

From Figure 1 the series has a clear increasing trend over the full training period. There is also mild seasonality / short-term monthly variation, but the dominant pattern is a smooth upwards evolution in the stock of registered vehicles.

2 Linear trend model

We consider a global linear trend model

$$Y_t = \theta_1 + \theta_2 x_t + \varepsilon_t, \qquad \varepsilon_t \sim \mathcal{N}(0, \sigma^2), \text{ i.i.d.} \tag{1}$$

where $t = 1, \ldots, N$ indexes the training observations.

2.1 Matrix form for the first 3 time points

For the first three time points, the model can be written in matrix notation as

$$\mathbf{y} = \mathbf{X}\boldsymbol{\theta} + \boldsymbol{\varepsilon}, \quad \mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}, \quad \mathbf{X} = \begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ 1 & x_3 \end{bmatrix}, \quad \boldsymbol{\theta} = \begin{bmatrix} \theta_1 \\ \theta_2 \end{bmatrix}.$$

$y_t = \theta_1 + \theta_2 \cdot x_t + \varepsilon_t \quad : \quad \varepsilon_t \sim N(0, \sigma^2) \quad (\text{iid})$

in the matrix form $\quad y = X\theta + \varepsilon$

output vector $\quad y = \begin{vmatrix} 2.930 \\ 2.934 \\ 2.941 \end{vmatrix}$

Design matrix $= \begin{vmatrix} 1 & 2018.000 \\ 1 & 2018.083 \\ 1 & 2018.163 \end{vmatrix}$

$$\Rightarrow \begin{vmatrix} 2.930 \\ 2.934 \\ 2.941 \end{vmatrix} = \begin{vmatrix} 1 & 2018.000 \\ 1 & 2018.083 \\ 1 & 2018.163 \end{vmatrix} \begin{vmatrix} \theta_1 \\ \theta_2 \end{vmatrix} + \begin{vmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \varepsilon_3 \end{vmatrix}$$

Figure 2: Handwritten derivation for the matrix form of the first three time points (Q2.1).

3 OLS - global linear trend model

3.1 Estimate the parameters θ_1 and θ_2 using the training set.

Given $\mathbf{y}$ and $\mathbf{X}$, the OLS estimator solves the normal equations:

$$\hat{\boldsymbol{\theta}}_{\text{OLS}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}.$$

Residuals are $\hat{\mathbf{e}} = \mathbf{y} - \mathbf{X}\hat{\boldsymbol{\theta}}$. The noise variance is estimated as $\hat{\sigma}^2 = \text{RSS}/(N-p)$ where $p = 2$ and $\text{RSS} = \hat{\mathbf{e}}^\top \hat{\mathbf{e}}$. The parameter covariance is then

$$\widehat{\text{Var}}(\hat{\boldsymbol{\theta}}) = \hat{\sigma}^2 (\mathbf{X}^\top \mathbf{X})^{-1},$$

and standard errors are the square roots of the diagonal entries.

3.2 Present the values of the parameter estimates θ_1 and θ_2 and their estimated standard errors $\hat{\sigma}_{\hat{\theta}_1}$ and $\hat{\sigma}_{\hat{\theta}_2}$. Plot the estimated mean as a line with the observations as points

Using the training set (2018-Jan to 2023-Dec, $N = 72$), the OLS estimates are

$$\hat{\theta}_1 = 2.944, \qquad \hat{\theta}_2 = 0.0561,$$

with estimated standard errors

$$\widehat{\sigma}_{\hat{\theta}_1} = 0.00610, \qquad \widehat{\sigma}_{\hat{\theta}_2} = 0.00178.$$

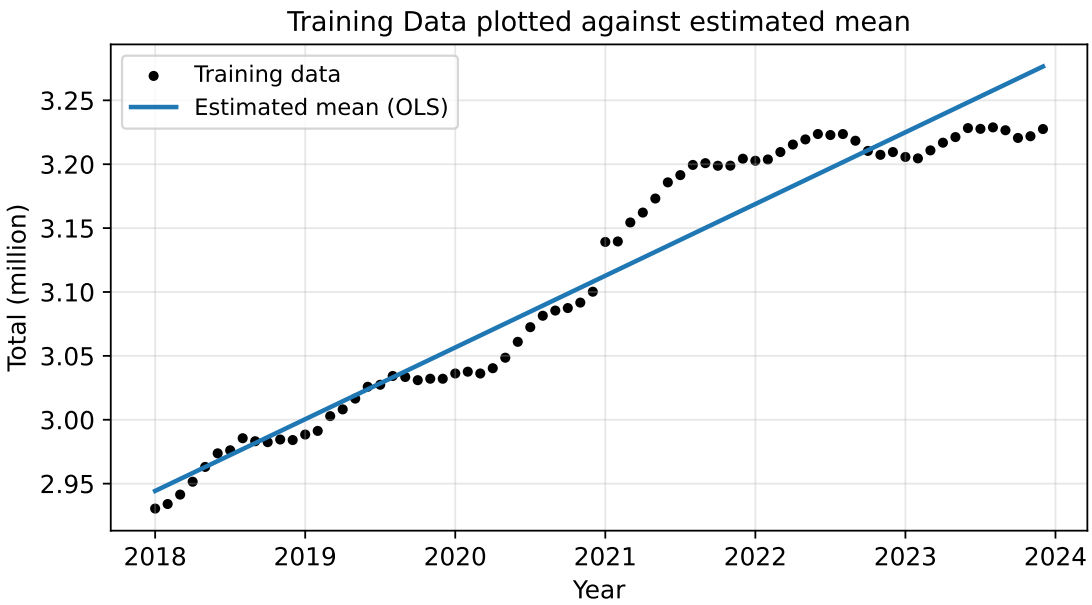

Figure 3: Training data plotted against estimated mean.

3.3 Make a forecast for the test set, hence the following 12 months

Table 1 presents the OLS point forecasts for the test period (2024-Jan to 2024-Dec).

Table 1: Estimated values $\hat{y}$ for each time point in 2024 for OLS.

Year-Month	Year	No. Vehicles (Million)
2024-01	6.000	3.281
2024-02	6.083	3.286
2024-03	6.167	3.291
2024-04	6.250	3.295
2024-05	6.333	3.300
2024-06	6.417	3.305
2024-07	6.500	3.309
2024-08	6.583	3.314
2024-09	6.667	3.319
2024-10	6.750	3.323
2024-11	6.833	3.328
2024-12	6.917	3.333

3.4 Plot the fitted model together with the training data and the forecasted values

Figure 4 shows the forecast together with the training data and the 95% prediction intervals.

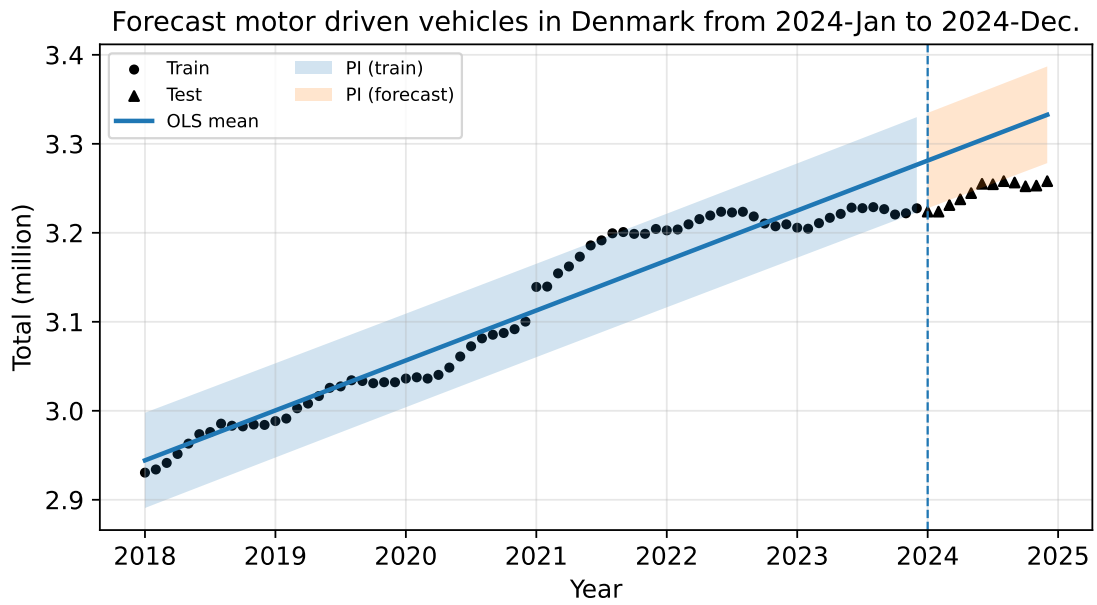

Figure 4: Forecasted values $\hat{y}$ and 95% prediction intervals for OLS.

3.5 Comment on forecast

The OLS model extrapolates a single global increasing trend. In this dataset the growth appears to slow down around the end of the training period, therefore the OLS forecast tends to slightly overpredict the levels during 2024.

3.6 Investigate the residuals of the model. Are the model assumptions fulfilled?

Figure 5 shows residuals over time and a Q–Q plot. A good fit would show residuals that are approximately centered around zero, with no clear time dependence, indicating a certain degree of autocorrelation, and a Q–Q plot close to a straight line that do not follow a normal distribution, , which does not satisfy the current assumption $\varepsilon_t \sim \mathcal{N}(0, \sigma^2)$

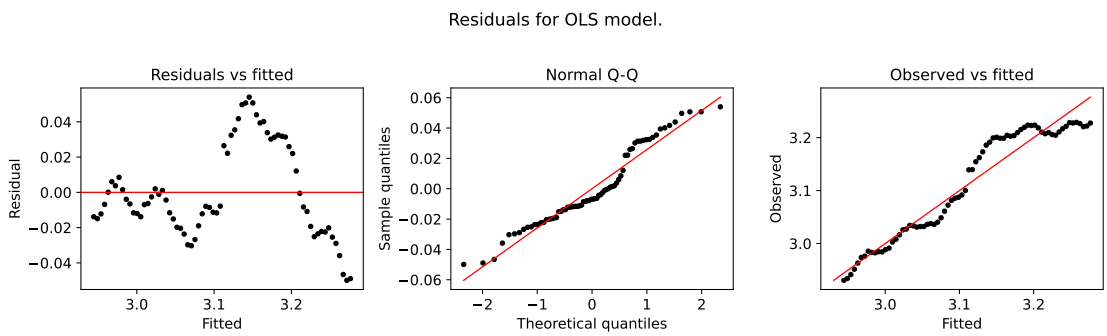

Figure 5: Residual analysis: residuals vs. time and Q–Q plot.

We also inspect autocorrelation and partial autocorrelation. Ideally, residuals should look like white noise. Figure 6 indicates some temporal dependence at short lags, suggesting that the i.i.d. noise assumption is not perfectly satisfied. In addition, the figure also shows a cutoff after lag 1 in PACF indicating an AR(1) process, suggesting that the current value of the series is based on the immediately preceding value.

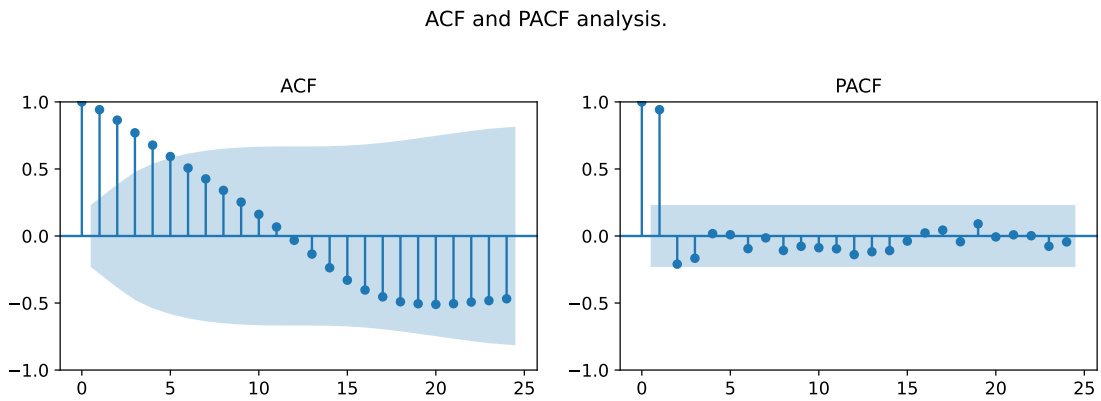

Figure 6: Autocorrelation function and partial autocorrelation function (train set residuals).

4 WLS - local linear trend model

In a local trend approach, newer observations are weighted more heavily. With forgetting factor $\lambda \in (0, 1)$, the weight of observation t (counting from the start of the training set) is

$$w_t = \lambda^{N-t},$$

so that $w_N = 1$ for the latest observation and weights decay exponentially into the past.

4.1 Variance-covariance matrix

For the global OLS model the noise covariance is $\Sigma_{\text{OLS}} = \sigma^2 \mathbf{I}$. In the WLS interpretation used here, the local model can be written with a diagonal covariance

$$\Sigma_{\text{WLS}} = \text{diag}\left(\frac{\sigma^2}{w_1}, \frac{\sigma^2}{w_2}, \ldots, \frac{\sigma^2}{w_N}\right),$$

which gives small variance (high precision) to recent points and large variance (low precision) to older points. Thus, compared to OLS, WLS allows the latest part of the series to dominate the fitted trend.

4.2 The λ-weights over time

Figure 7 visualizes how the training observations are weighted when $\lambda = 0.9$. The highest weight is assigned to the most recent training point (2023-Dec).

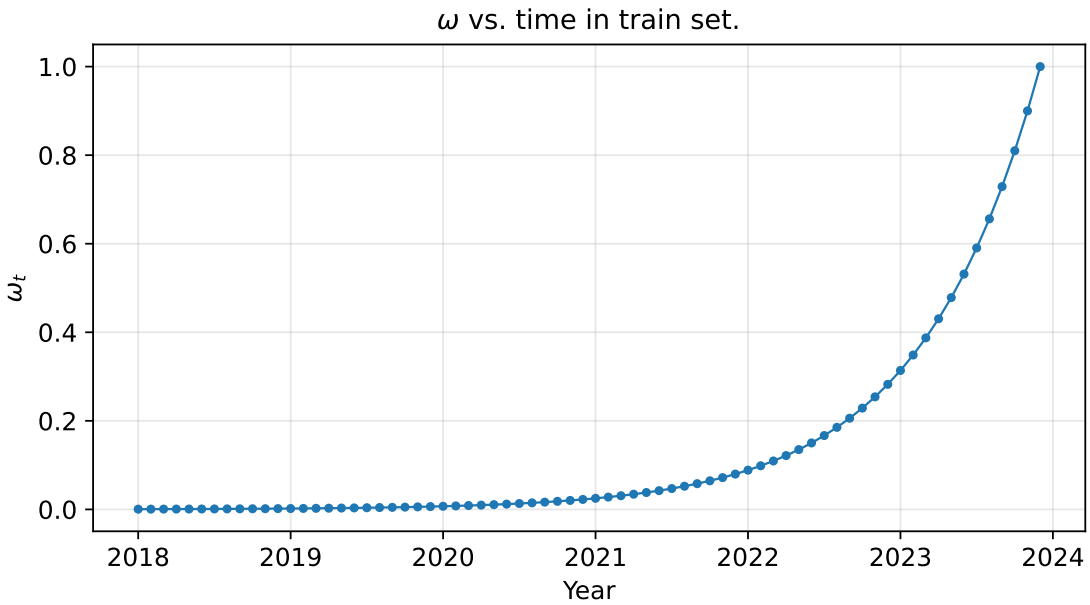

Figure 7: λ-weights for WLS model with $\lambda = 0.9$.

4.3 4.3 Compare the sum of weights (ω_i) in WLS and OLS

In the local (WLS) trend model, the training observations are not treated equally. Instead, we apply exponentially decaying "memory" weights such that the most recent observation receives the largest weight and older observations are down-weighted. Specifically, let the weights be

$$\omega_j = \lambda^j, \qquad j = 0, 1, \ldots, N-1,$$

where $j = 0$ corresponds to the most recent observation, $j = 1$ to the second most recent, etc. The total (effective) weight is therefore the geometric sum

$$T \;=\; \sum_{j=0}^{N-1} \omega_j \;=\; \sum_{j=0}^{N-1} \lambda^j \;=\; \frac{1-\lambda^N}{1-\lambda}, \qquad 0 < \lambda < 1.$$

For $\lambda = 0.9$ and $N = 72$, we obtain

$$T \approx 9.994925,$$

which is far smaller than the actual number of observations. This highlights an important interpretation: although we still have $N = 72$ months available, the exponentially decaying weights make the model behave as if it had only about $T \approx 10$ "effective" observations contributing substantially to the fit.

In contrast, for the global OLS model all observations have equal weight $\omega_i = 1$, giving

$$T_{\text{OLS}} = \sum_{i=1}^{N} 1 = N = 72.$$

Thus, OLS uses the full information content of all training points equally, while WLS intentionally reduces the influence of older observations in order to adapt to potential changes in the underlying trend.

This difference in effective information also affects the variance estimator in WLS. Using the GLS/WLS formulation, the error variance is estimated as

$$\hat{\sigma}^2 \;=\; \frac{(\mathbf{Y} - \mathbf{X}\hat{\boldsymbol{\theta}})^\top \boldsymbol{\Sigma}^{-1} (\mathbf{Y} - \mathbf{X}\hat{\boldsymbol{\theta}})}{T - p},$$

where $\mathbf{Y}$ is the observation vector, $\mathbf{X}$ is the design matrix, $\hat{\boldsymbol{\theta}}$ is the parameter estimate, and p is the number of parameters. Because $T \ll N$ in the local model, the denominator $T-p$ is smaller than the OLS denominator $N-p$, which typically leads to a larger estimated error variance (all else equal). Intuitively, the model "counts" fewer effective observations because older data points are heavily down-weighted, reducing the overall informational content used to estimate the parameters.

Finally, the requirement

$$T > p$$

provides a practical lower bound on λ: we must have enough effective observations to reliably estimate the p parameters. If λ is chosen too small, T can become close to p (or even below), making the variance estimate unstable and the parameter estimation unreliable.

4.4 Estimate and present $\hat{\theta}_1$ and $\hat{\theta}_2$ corresponding to the WLS model with $\lambda = 0.9$

The WLS estimator is

$$\hat{\boldsymbol{\theta}}_{\text{WLS}} = (\mathbf{X}^\top \Sigma^{-1} \mathbf{X})^{-1} \mathbf{X}^\top \Sigma^{-1} \mathbf{y}.$$

Table 2 compares the OLS and WLS parameter estimates for $\lambda = 0.9$.

Table 2: Estimated parameters $\hat{\theta}_1$ and $\hat{\theta}_2$ for OLS and WLS models with $\lambda = 0.9$.

Model	$\hat{\theta}_1$	$\hat{\theta}_2$
OLS	2.9443	0.0561
WLS ($\lambda = 0.9$)	3.0725	0.0275

4.5 Make a forecast for the next 12 months - i.e., compute predicted values corresponding to the WLS model with $\lambda = 0.9$

Figure 8 compares OLS and WLS forecasts. Since WLS gives more emphasis to the latest training observations, it produces a smaller slope and therefore a flatter forecast, which matches the 2024 test values better than the global OLS trend.

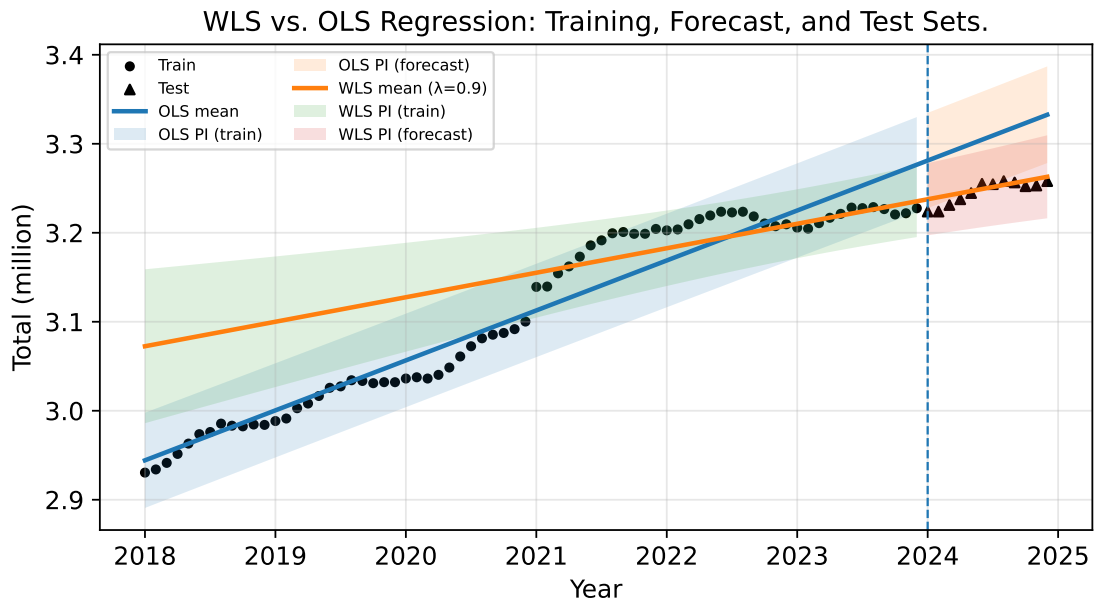

Figure 8: WLS and OLS predictions for the next 12 months (2024).

When it comes to predictive accuracy, the WLS model performs significantly better than the OLS model. The Mean Absolute Error (MAE) for WLS sits at roughly 0.00688, compared to 0.06117 for OLS, meaning the WLS model is about 8.9 times more accurate on average. Additionally, both the Root Mean Squared Error (RMSE) and Relative Absolute Error (RAE) are noticeably lower for the WLS model.

The WLS model is the clear choice over the OLS model. Because WLS accounts for the recentness of the data, it enhances test set predictions significantly, reducing the average error by a factor of almost nine and proving to be a vastly more powerful forecasting tool.

4.6 Effect of λ on the forecast (optional)

Decreasing λ makes the model more "local" (more reactive), and therefore the estimated slope tends to follow the most recent changes more closely.

5 Recursive estimation and optimization of λ

5.1 Update equations for R_t and $\hat{\theta}_t$

Let $x_t^\top = [1 \;\; x_t]$ denote the regressor row for time t. Without forgetting, the recursive matrices are

$$R_t = R_{t-1} + x_t x_t^\top, \qquad \hat{\theta}_t = \hat{\theta}_{t-1} + R_t^{-1} x_t \, (y_t - x_t^\top \hat{\theta}_{t-1}).$$

With forgetting factor λ, the R_t recursion becomes

$$R_t = \lambda R_{t-1} + x_t x_t^\top,$$

(and the same update structure for $\hat{\theta}_t$).

The first two iterations for R_t (up to R_2), starting from $R_0 = 0.1\, I$ and $\theta_0 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$, are included.

5.1) Step 1 ($t=1$): $X_1 = |1, 0|$ $R_0 = \begin{vmatrix} 0.1 & 0 \\ 0 & 0.1 \end{vmatrix}$

$$R_1 = \lambda(1) R_0 + X_1^T X_1$$

without forgetting ($\lambda = 1$)

$$R_1 = 1 \cdot \begin{vmatrix} 0.1 & 0 \\ 0 & 0.1 \end{vmatrix} + \begin{vmatrix} 1 \\ 0 \end{vmatrix} |1 \quad 0|$$

$$= \begin{vmatrix} 0.1 & 0 \\ 0 & 0.1 \end{vmatrix} + \begin{vmatrix} 1 & 0 \\ 0 & 0 \end{vmatrix} = \begin{vmatrix} 1.1 & 0 \\ 0 & 0.1 \end{vmatrix}$$

$$\hat{\theta}_1 = R_1^{-1} h_1 \qquad h_1 = \begin{vmatrix} 2.930 \\ 0 \end{vmatrix}$$

$$\hat{\theta}_1 = R_1^{-1} \begin{vmatrix} 2.930 \\ 0 \end{vmatrix} = \begin{vmatrix} 2.664 \\ 0 \end{vmatrix}$$

Step 2 ($t=2$): $X_2 = |1 \quad 0.083|$

$$R_2 = R_1 + X_2^T X_2$$

$$R_2 = \begin{vmatrix} 1.1 & 0 \\ 0 & 0.1 \end{vmatrix} + \begin{vmatrix} 1 \\ 0.083 \end{vmatrix} |1 \quad 0.083|$$

$$= \begin{vmatrix} 1.1 & 0 \\ 0 & 0.1 \end{vmatrix} + \begin{vmatrix} 1 & 0.083 \\ 0.083 & 0.083^2 \end{vmatrix} = \begin{vmatrix} 2.1 & 0.083 \\ 0.083 & 0.1069 \end{vmatrix}$$

$$\hat{\theta}_2 = R_2^{-1} h_2 \quad : \quad h_2 = \begin{vmatrix} 5.885 \\ 0.245 \end{vmatrix}$$

$$\hat{\theta}_2 = R_2^{-1} \begin{vmatrix} 5.885 \\ 0.245 \end{vmatrix} = \begin{vmatrix} 2.78 \\ 0.119 \end{vmatrix}$$

Figure 9: Handwritten derivation for the first two iterations of R_t (Q5.1).

5.2 Implementation and estimates up to $t = 3$

The recursive formulas are implemented in a simple loop. Table 3 shows the first available parameter estimates and one-step predictions.

Table 3: Recursive estimates $\hat{\theta}_{1,t}$ and $\hat{\theta}_{2,t}$ (and one-step prediction $\hat{y}_{t|t-1}$) up to $t = 3$.

t	$\hat{\theta}_{1,t}$	$\hat{\theta}_{2,t}$	$\hat{y}_{t\|t-1}$
1	–	–	–
2	2.930	0.043	–
3	2.930	0.066	2.938
4	–	–	2.946

Even though the computations are matrix-based, the logic is intuitive: at each new observation we update the information matrix R_t, then adjust $\hat{\theta}_t$ in the direction that reduces the latest prediction error.

5.3 RLS estimates at time $t = N$ compared to OLS

For $\lambda = 1$ and consistent initialization, the recursive least squares solution converges to (and in practice matches) the batch OLS solution computed from the same data. Small differences can occur if the initial R_0 and θ_0 are not chosen carefully, since the first updates can dominate the early trajectory.

5.4 RLS with forgetting: parameter trajectories

Figure 10 shows the evolution of (θ_1, θ_2) for $\lambda = 1$. Figure 11 compares parameter trajectories for $\lambda = 0.7$ and $\lambda = 0.99$. Smaller λ reacts more strongly to recent data (more variability), while larger λ is closer to the global trend (more stable).

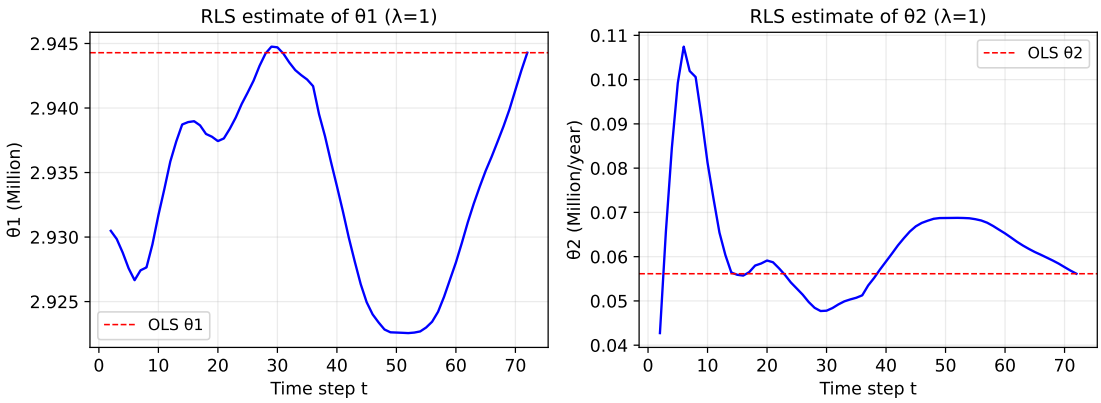

Figure 10: Iterative evolution of RLS regression coefficients: intercept (θ_1) and slope (θ_2) for $\lambda = 1$.

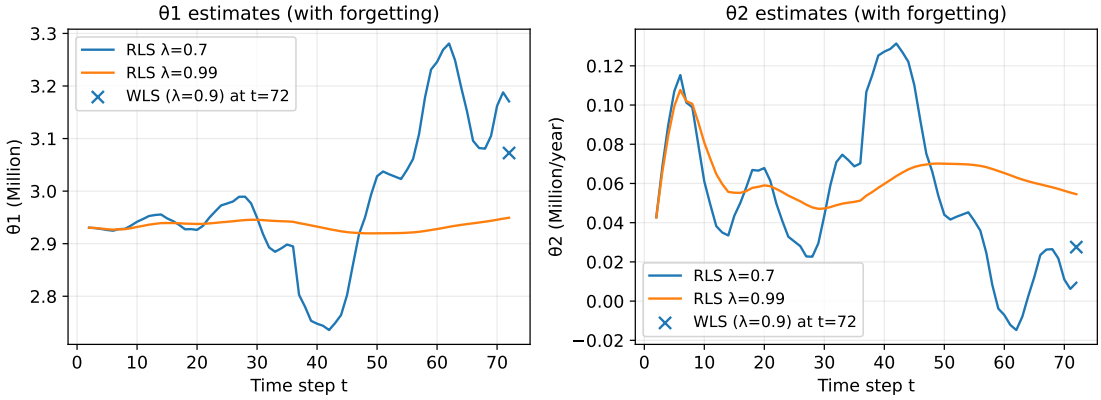

Figure 11: Iterative evolution of RLS regression coefficients: intercept and slope for $\lambda = 0.7$ and $\lambda = 0.99$.

5.5 One-step predictions and residuals

One-step predictions are computed as

$$\hat{y}_{t+1|t} = x_{t+1}^{\top}\hat{\theta}_t, \qquad \hat{\varepsilon}_{t+1|t} = \hat{y}_{t+1|t} - y_{t+1}.$$

Figure 12 shows the one-step predictions for $\lambda = 0.7$ and $\lambda = 0.99$. Figure 13 shows the corresponding residuals (after a short burn-in period).

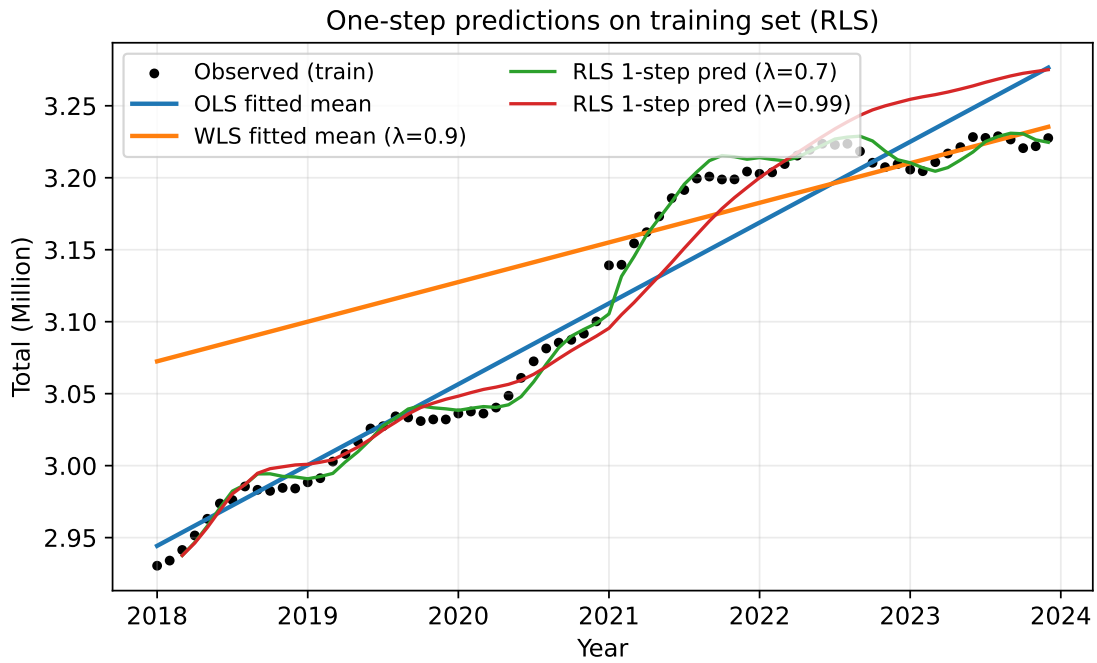

Figure 12: One-step predictions of the whole dataset for $\lambda = 0.7$ and $\lambda = 0.99$.

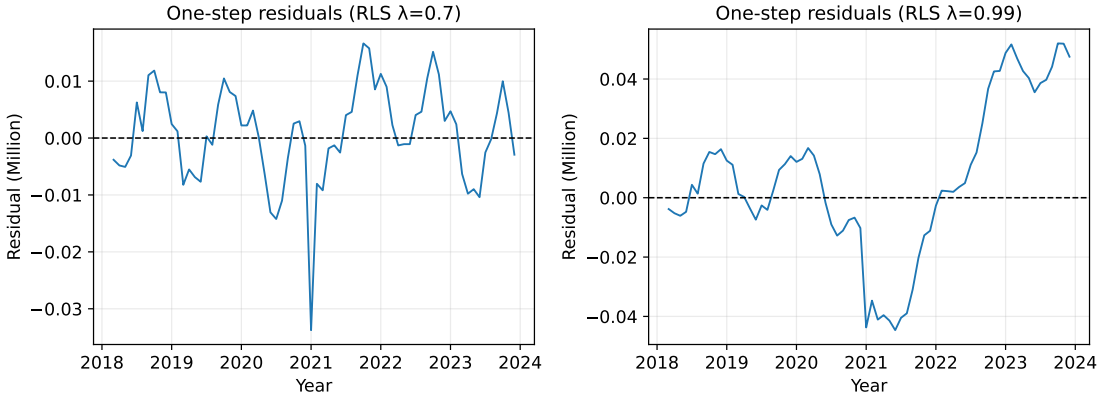

Figure 13: One-step residuals of the whole dataset for $\lambda = 0.7$ and $\lambda = 0.99$.

5.6 Optimizing λ for horizons $k = 1, \ldots, 12$

For each horizon k, we compute the k-step residuals

$$\hat{\varepsilon}_{t+k|t} = \hat{y}_{t+k|t} - y_{t+k},$$

and the corresponding RMSE_k over the available time points. Figure 14 plots RMSE as a function of λ for horizons $k = 1, \ldots, 12$.

In general, shorter horizons often prefer smaller λ (more adaptivity), while longer horizons tend to prefer larger λ (more stability). Therefore, it can be reasonable to let the optimal λ depend on the forecast horizon.

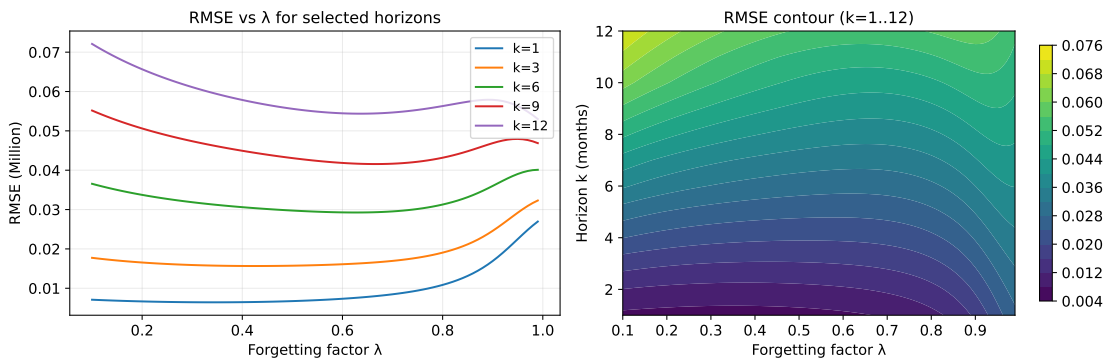

Figure 14: RMSE values for different horizons and forgetting factors λ.

5.7 Predictions of the test set using RLS

Figure 15 compares forecasts for 2024 obtained from OLS, WLS, and an RLS-based approach. Since the series shows a slowdown in growth near the end of the training window, the adaptive methods (WLS/RLS) are typically closer to the test set than the global OLS trend.

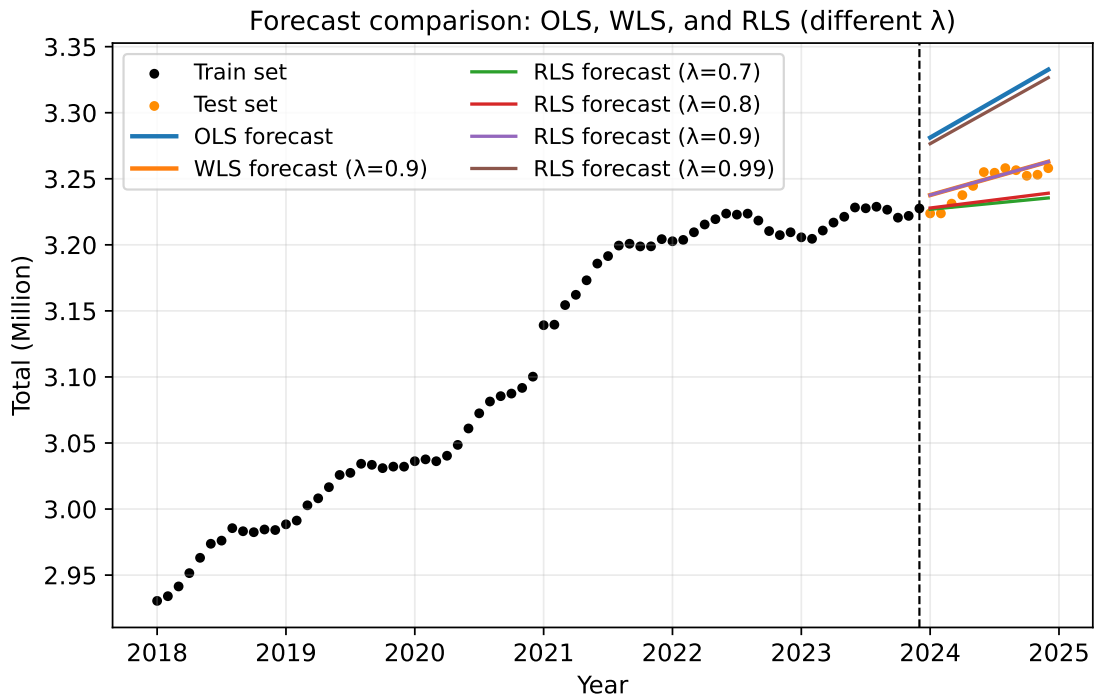

Figure 15: Forecast comparison of OLS, WLS and RLS (2024).

5.8 Reflections on time-adaptive models

Time-adaptive models can reduce bias when the underlying trend changes, but they also introduce new pitfalls:

- **Overfitting vs. underfitting:** a very small λ can overreact to noise (high variance), while a very large λ can be too rigid (high bias).
- **Test set design:** temporal dependence means we cannot randomly shuffle observations. A realistic evaluation must respect time ordering.
- **Recursive prediction benefits:** RLS allows continuous updating and can be useful in practice when new data arrives, but it does not remove the need for careful out-of-sample testing.
- **Other adaptive techniques:** rolling-window regression, state-space models (e.g. local level / local trend), and Kalman filtering provide principled alternatives for time-varying parameters.

Use of Generative AI

The generative AI tool I used was ChatGPT (OpenAI, GPT-5.1). It was used to help brainstorm the assignment , structure the jupyter notebook, and support the transition of parts of the workflow from R to Python in order to improve reproducibility and efficiency.

I reviewed, adapted, and tested all suggested code before using it, and I edited it to match my own understanding and the actual results.

All final design decisions, experiments, and interpretations are my own responsibility.

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